

① A coil of area of cross section 0.5 m^2 is placed in a magnetic field acting normally to its plane. The field varies as $B = 0.5t^2 + 2t$, where B is in Tesla and t in seconds. The emf induced in the coil at $t=1$ is : A) 0.5 V B) 1.0 V C) 1.5 V D) 3.0 V

② In a young's double slit experiment in air, the fringe width is found to be 0.44 mm . If the entire setup is immersed in water $\mu = \frac{4}{3}$, the fringe width will be ? A) 0.88 mm B) 0.44 mm C) 0.33 mm D) 0.59 mm

③ Calculate the energy released/absorbed in Mev in the following nuclear reaction?



Given $m({}^1_1\text{H}) = 1.007825 \text{ u}$

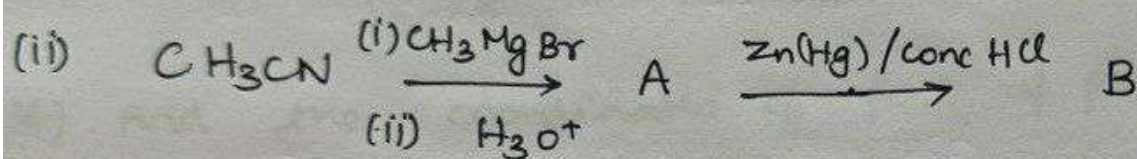
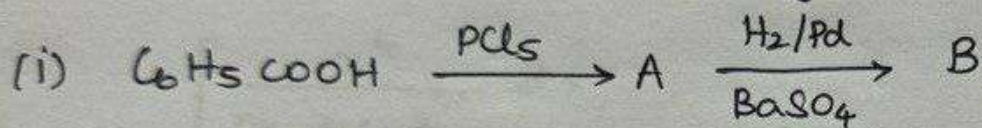
$$m({}^2_1\text{H}) = 2.014102 \text{ u}$$

$$m({}^3_1\text{H}) = 3.016049 \text{ u}$$

④ A photosensitive surface of work function 2.1 eV is irradiated by radiation of wavelength 150 nm . Calculate (i) threshold wavelength, (ii) Energy (in eV) of incident photon and (iii) Maximum KE of the emitted photoelectron.

⑤ When FeCr_2O_4 is fused with Na_2CO_3 in the presence of air, it gives a yellow solution of compound A. Compound A on acidification gives compound B. Compound B on reaction with KCl forms an orange coloured compound C. Compound C acidified solution oxidises Na_2SO_3 to D. Identify A, B, C, D.

⑥ Write structures of main compounds A and B in each of the following reactions?



⑦ Define the following terms:

(i) Invert Sugar

(ii) Native Protein

(iii) Nucleotide

⑧ Although both $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ have sp^3 hybridisation yet $[\text{NiCl}_4]^{2-}$ is paramagnetic and $\text{Ni}(\text{CO})_4$ is diamagnetic. Give reason.

⑨ Write the electronic configuration of d^5 on the basis of crystal field theory when
(i) $\Delta_0 < P$ and (ii) $\Delta_0 > P$.

10) The sum of the order and degree of the differential equation $\frac{d}{dx} \left(\left(\frac{dy}{dx} \right)^3 \right)$.

- a) 2 b) 3 c) 5 d) 0

11) Find the vector and cartesian equations of a line that passes through the point $A(1, 2, -1)$ and parallel to the line $5x - 2y = 14 - 7z$.

12) Differentiate $\sec^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right)$ w.r.t $\sin(2x\sqrt{1-x^2})$

13) $\int \frac{x^4}{(x-1)(x^2+1)} dx$

14) Find the coordinates of foot of perpendicular drawn from the point $P(0, 2, 3)$ to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$.

15) If $x = a \cos t$ and $y = b \sin t$, then find $\frac{d^2y}{dx^2}$.

16) If $|A| = |kA|$, where A is a square matrix of order 2, then sum of all possible values of k is?

17) If $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 3 & a & 1 \end{bmatrix}$ is non singular and a GA,

then what is set A ?